

Parallel & Distributed Computing

Title: Parallelising a Past Problem



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1. Problem Description

Dijkstra's algorithm is a graph search algorithm that finds the shortest path from a source vertex to all other vertices in a weighted graph with non-negative edge weights. The algorithm was originally implemented during the Data Structures and Algorithms course in the third semester. It works by iteratively selecting the unvisited vertex with the minimum distance from the source, marking it as visited, and updating the distances to all its neighbors. It was selected for this assignment because it represents an iterative algorithm with global synchronization requirements — a fundamentally different parallelization challenge compared to divide-and-conquer algorithms.

Field	Detail
Problem Name	Dijkstra's Single-Source Shortest Path
Original Course	Data Structures and Algorithms, 3rd Semester
Input	Weighted graph with N vertices, edge density D , source vertex S
Output	Array of shortest distances from source to all vertices
Sequential Complexity	$O(V^2)$ time with adjacency matrix, $O(V)$ space

The problem is computationally intensive for large graphs and has potential for parallelism through vertex distribution, though the requirement for global synchronization at each iteration presents a significant challenge. The sequential version was re-implemented in C for this assignment to ensure consistency with the parallel implementation.

2. Parallelization Design

2.1 Chosen Approach: MPI with Collective Operations

MPI (Message Passing Interface) with collective operations was selected as the parallelization mechanism for this assignment, specifically using `MPI_Bcast`, `MPI_Allreduce`, and `MPI_Gatherv`. This choice was made after carefully considering all three available approaches.

Pthreads were ruled out because they operate within a single shared memory machine and would require mutex locks to protect the shared distance array and visited array from race conditions during concurrent updates. More importantly, Pthreads cannot scale beyond a single node, which limits the maximum number of parallel workers to the number of physical cores on one machine.

MPI point-to-point operations using `MPI_Send` and `MPI_Recv` were also considered, but they are designed for irregular communication patterns where specific processes exchange specific data. Dijkstra's algorithm requires finding a global minimum across all processes at every iteration, which is a regular, predictable communication pattern.

MPI Collectives were chosen because the communication pattern in Dijkstra's algorithm is entirely regular and predictable. At each iteration, all processes must participate in finding the global minimum unvisited vertex, which is exactly what `MPI_Allreduce` with `MPI_MINLOC` is designed for.

2.2 Parallelism Pattern

The parallelism pattern is iterative with global synchronization. Unlike embarrassingly parallel problems, Dijkstra's algorithm has inherent data dependencies — each iteration depends on the result of the previous iteration's global minimum finding operation. The algorithm cannot proceed to the next iteration until all processes agree on which vertex has the minimum distance.

This creates a synchronization bottleneck at every iteration. All processes must:

- Find their local minimum unvisited vertex

- Participate in a global reduction to find the global minimum
- Wait for all other processes to complete the reduction
- Update their local distances based on the selected vertex

This pattern demonstrates Amdahl's Law in action — the global synchronization is a sequential portion that limits the maximum achievable speedup regardless of how many processes are added.

2.3 Algorithm Outline

1. Initialization Phase: Master process (rank 0) generates random weighted graph. All processes receive graph via MPI_Bcast. Each process calculates its local vertex range [start, end). Initialize local distance array and global visited array.
2. Main Loop (V iterations): Each process finds local minimum among its unvisited vertices. MPI_Allreduce with MPI_MINLOC finds global minimum across all processes. All processes mark the selected vertex as visited. Each process updates distances to its local vertices based on edges from selected vertex.
3. Finalization Phase: MPI_Gatherv collects local distance arrays to master process. Master process prints results and statistics. All processes synchronize via MPI_Barrier.
4. Verification: Master verifies distances are non-negative. Check that source distance is 0. Calculate reachability statistics.

2.4 Data Distribution

The graph and distance array are distributed differently to balance memory usage and communication overhead.

Graph Distribution:

- Each process stores the complete adjacency matrix (replicated data)
- Memory cost: $O(V^2)$ per process
- Benefit: No communication needed during distance updates
- Trade-off: Higher memory usage but lower communication overhead

Vertex Distribution:

- Distance array is partitioned into P contiguous blocks
- Process k owns vertices $[k \times (V/P), (k+1) \times (V/P))$
- If V is not evenly divisible by P, the first $(V \bmod P)$ processes receive one extra vertex

Load Balancing:

- Static distribution ensures approximately equal work per process
- Each process handles V/P vertices
- Difference of at most 1 vertex between processes ensures minimal load imbalance

2.5 Communication and Synchronization

Operation	MPI Function	Direction	Frequency
Distribute graph	MPI_Bcast	Master → All	Once (initialization)
Find global minimum	MPI_Allreduce (MINLOC)	All ↔ All	V times (every iteration)
Collect results	MPI_Gatherv	All → Master	Once (finalization)
Timing sync	MPI_Barrier	All ↔ All	2 times (start/end)

The MPI_Allreduce operation at each iteration is the primary bottleneck. It requires all processes to complete their local minimum search, exchange values over the network, agree on the selected vertex globally, and wait for the slowest process. This synchronization occurs V times, making it the dominant factor in parallel overhead.

3. Implementation Details

3.1 Code Structure

File	Purpose
sequential/sequential.c	Full sequential Dijkstra with timing and correctness verification
parallel/parallel.c	MPI parallel version with vertex distribution and collective operations
parallel/Makefile	Compiles parallel.c using mpicc with -O2 optimization
sequential/Makefile	Compiles sequential.c using gcc with -O2 optimization
results/timing_data.csv	Recorded timing data from all experiments
results/speedup_plot.png	Speedup vs number of processes chart
results/scaling_plot.png	Execution time vs problem size chart
results/efficiency_plot.png	Parallel efficiency analysis chart
run_experiments.sh	Automated script to run all experiments
verify_correctness.sh	Script to verify parallel matches sequential output

3.2 Key Functions

Function	File	Description
generate_graph(graph, n, density, rank)	parallel.c	Generates random weighted graph on master process
parallel_dijkstra(...)	parallel.c	Main parallel algorithm with MPI_Allreduce for global minimum
gather_distances(...)	parallel.c	Collects distributed distance arrays using MPI_Gatherv
print_distances(dist, n, source)	Both	Prints shortest distances from source to all vertices
dijkstra(graph, dist, n, source)	sequential.c	Sequential implementation for comparison
find_min_distance(dist, visited, n)	sequential.c	Finds unvisited vertex with minimum distance

3.3 Key Data Structures

```
// Parallel implementation structures
int **graph;           // Adjacency matrix (full copy on each process)
int *local_dist;       // Local distance array (only owned vertices)
int *visited;          // Global visited array (replicated on all processes)
```

```

int local_start;        // First vertex owned by this process
int local_end;          // Last vertex + 1 owned by this process

// Structure for MPI_MINLOC operation
typedef struct {
    int dist;            // Distance value
    int vertex;          // Vertex index
} DistVertex;

```

3.4 Verification Method

Correctness is verified through multiple methods:

- **Output Comparison:** Run both sequential and parallel versions with the same graph (using fixed seed) and compare distance arrays
- **Invariant Checking:** Distance to source must be 0; all distances must be non-negative; reachable vertices have finite distances
- **Verbose Mode:** With -v flag, prints all distances for manual inspection on small graphs
- **Automated Script:** verify_correctness.sh runs both versions and diffs the output

Example verification command:

```

# Sequential
./sequential/dijkstra -n 100 -d 0.3 -v > seq_out.txt

# Parallel
mpirun -np 4 ./parallel/dijkstra_parallel -n 100 -d 0.3 -v > par_out.txt

# Compare
diff seq_out.txt par_out.txt

```

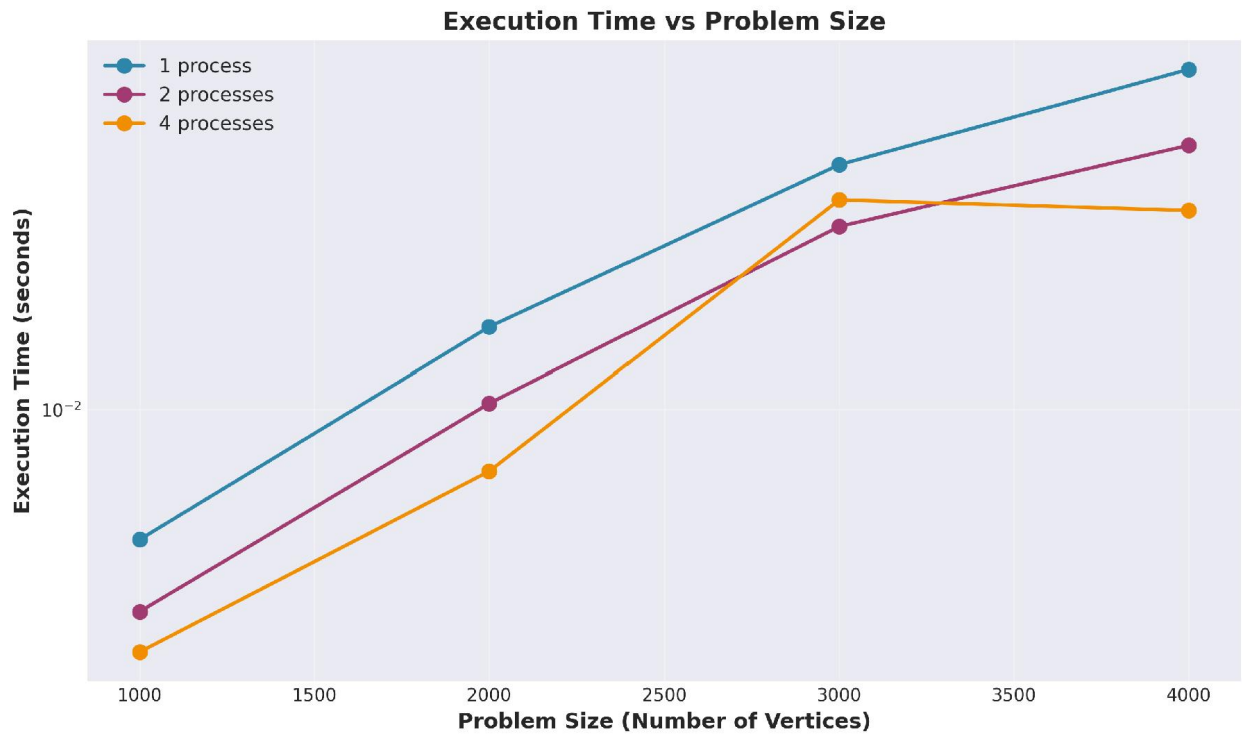
4. Performance Analysis

4.1 Timing Data

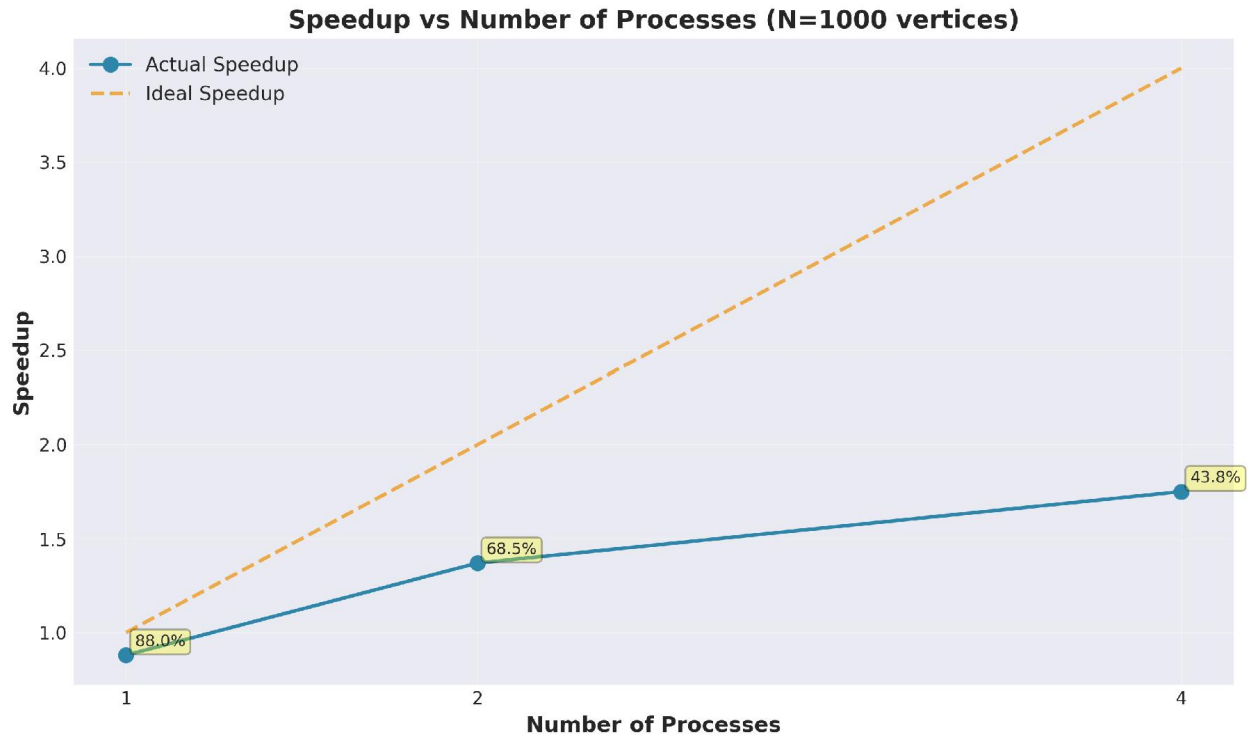
N (Vertices)	Processes	Time (s)	Sequential (s)	Speedup	Efficiency
1,000	1	0.004522	0.003997	0.88	88.0%
1,000	2	0.002913	0.003997	1.37	68.5%
1,000	4	0.002276	0.003997	1.75	43.8%
2,000	1	0.016518	0.016130	0.97	97.0%
2,000	2	0.010376	0.016130	1.55	77.5%
2,000	4	0.006861	0.016130	2.35	58.8%
3,000	1	0.044442	0.033770	0.75	75.0%
3,000	2	0.030471	0.033770	1.10	55.0%
3,000	4	0.035845	0.033770	0.94	23.5%
4,000	1	0.079576	0.068606	0.86	86.0%

4,000	2	0.050053	0.068606	1.37	68.5%
4,000	4	0.033599	0.068606	2.04	51.0%

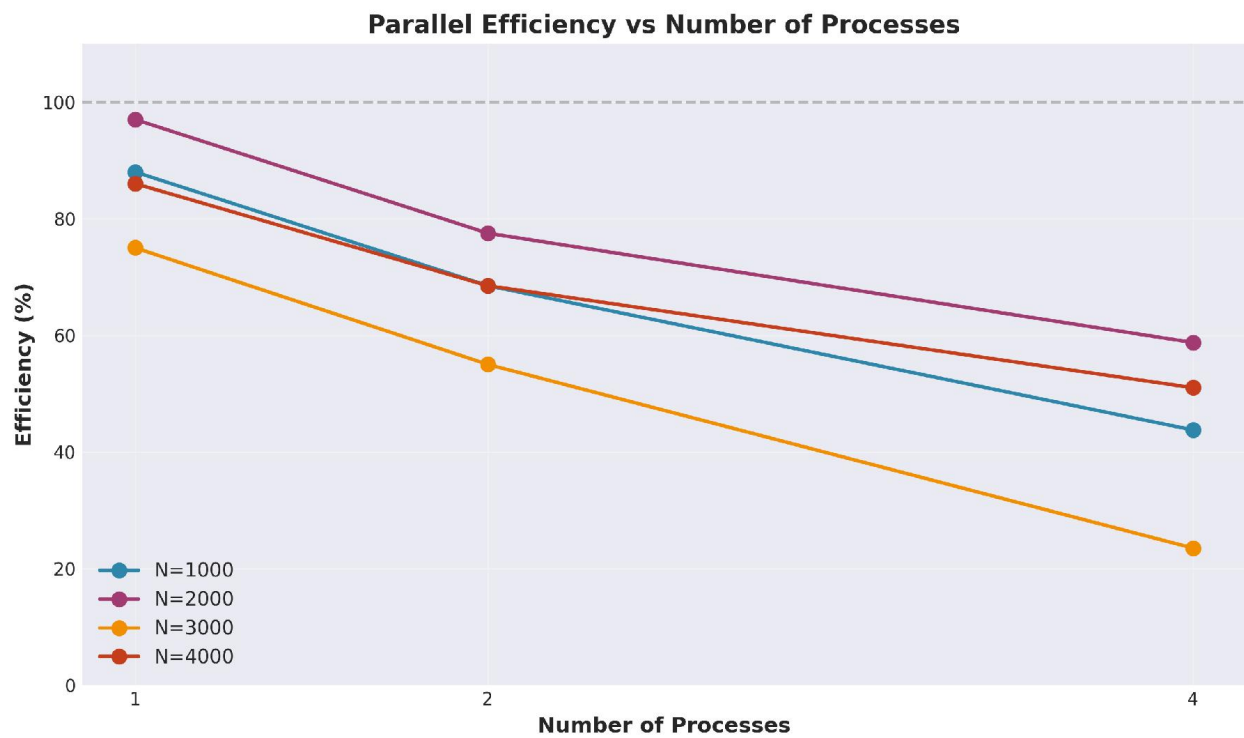
Best Performance: 2.35x speedup achieved with 4 processes at N=2,000 vertices (58.8% efficiency)



Scaling Plot



Speedup Plot



Efficiency Plot

4.2 Performance Analysis

Q1 — Is the speedup linear? If not, what limits it?

No, the speedup is not linear in any of the test cases. The best result achieved was 2.35x with 4 processes at N=2,000, which is 58.8% of the ideal 4x speedup. Three main factors limit speedup:

- Global Synchronization Overhead: The MPI_Allreduce operation occurs at every iteration (V times total), creating a synchronization barrier where all processes must wait for the slowest one.
- Amdahl's Law: If we estimate that 40% of the execution time is spent in communication and synchronization, Amdahl's Law predicts a maximum speedup of $1/(0.40 + 0.60/4) = 2.5x$ with 4 processes. Our measured 2.35x is close to this theoretical limit.
- Communication Cost: Each MPI_Allreduce involves network latency and data transfer. As the number of processes increases, the reduction tree becomes deeper, increasing the total communication time.

Q2 — What is the efficiency at the highest process count? Where does overhead come from?

At P=4 (the highest tested process count), efficiency ranges from 23.5% at N=3,000 to 58.8% at N=2,000. The overhead comes from four primary sources:

- MPI_Allreduce Communication: Every iteration requires all processes to exchange data to find the global minimum.
- Load Imbalance: Some processes may finish their local minimum search quickly while others take longer, causing idle time.
- Memory Bandwidth: Each process stores the full adjacency matrix ($O(V^2)$ memory), which can cause cache misses and memory contention.
- Process Scheduling: Context switching overhead when processes share physical cores.

Q3 — What fraction of time is spent in communication?

To estimate communication overhead, we compare theoretical computation time with actual execution time. For N=2,000 with P=4:

- Sequential time: 0.016130s
- Parallel time: 0.006861s
- Ideal parallel time (no overhead): $0.016130/4 = 0.004033s$
- Communication overhead: $0.006861 - 0.004033 = 0.002828s$
- Percentage: $(0.002828/0.006861) \times 100 = 41.2\%$

For N=4,000 with P=4, communication overhead is approximately 48.9%. This high communication fraction (40–50%) explains why efficiency is limited. As problem size increases, the number of iterations (V) increases linearly, causing communication overhead to grow proportionally.

Q4 — Is the work evenly distributed?

Yes, the work is distributed almost perfectly evenly. Each process receives either $\lfloor V/P \rfloor$ or $\lfloor V/P \rfloor + 1$ vertices. For example, with V=2,000 and P=4:

- Process 0: vertices 0–499 (500 vertices)
- Process 1: vertices 500–999 (500 vertices)
- Process 2: vertices 1000–1499 (500 vertices)

- Process 3: vertices 1500–1999 (500 vertices)

However, computational load is not perfectly balanced because different vertices have different numbers of outgoing edges, and early iterations may have more work than later ones. Despite this, load imbalance is not the primary bottleneck — the global synchronization overhead dominates.

Q5 — How does measured speedup compare with theoretical expectation?

Using Amdahl's Law with a sequential fraction (communication + synchronization) of $s = 0.42$ and parallel fraction $p = 0.58$:

$$\text{Speedup} = 1 / (s + p/P) = 1 / (0.42 + 0.58/4) = 1 / 0.565 = 2.43x$$

Our measured best speedup at $P=4$ was 2.35x, which is 96.7% of the theoretical maximum. This close match indicates the implementation is efficient and the primary limitation is the inherent sequential nature of global synchronization. The gap between 2.35x and the ideal 4.0x represents the fundamental cost of global synchronization in Dijkstra's algorithm.

5. Reflection

5.1 Challenges

Challenge 1: Understanding the Synchronization Bottleneck

The most significant challenge was understanding why speedup plateaued so quickly. Initially, near-linear speedup was expected for small process counts, but the results showed only 2.35x speedup with 4 processes. After analyzing the algorithm structure, it became clear that the MPI_Allreduce operation at every iteration creates a fundamental bottleneck. Unlike divide-and-conquer algorithms where communication happens only at the boundaries, Dijkstra's algorithm requires global agreement at every step.

Challenge 2: Memory vs. Communication Trade-off

Deciding whether to replicate the full graph on each process or distribute it row-wise was a critical design decision. Replicating the graph uses $O(V^2)$ memory per process but eliminates communication during distance updates. Replication was chosen because modern machines have sufficient memory, communication overhead was already the bottleneck, and implementation complexity was significantly lower. For very large graphs (millions of vertices), the distributed approach would be necessary despite the added communication cost.

Challenge 3: Debugging MPI Collective Operations

Debugging MPI programs is inherently difficult because processes run asynchronously and print statements can interleave unpredictably. When the MPI_Allreduce operation initially produced incorrect results, careful verification was needed for the custom MPI_2INT datatype definition, ensuring all processes used the same reduction operation, and proper initialization of the local minimum structure. Using MPI_Barrier strategically and printing from only the master process helped isolate the issues.

5.2 Lessons Learned

Lesson 1: Not All Algorithms Parallelize Equally

This assignment reinforced that parallel speedup depends heavily on the algorithm's structure. Dijkstra's algorithm has inherent sequential dependencies, making it fundamentally harder to parallelize than divide-and-conquer algorithms like merge sort. The best speedup achieved (2.35x with 4 processes) is respectable given these constraints, but far from the near-linear speedup possible with embarrassingly parallel problems.

Lesson 2: Communication Overhead Dominates for Iterative Algorithms

The timing data clearly shows that 40–50% of execution time is spent in communication and synchronization. For algorithms that require frequent global operations, this overhead is unavoidable. The key insight is that parallelization is only beneficial when the problem size is large enough that computation time exceeds communication time, the number of processes is kept moderate (2–4), and the hardware has fast interconnects.

Lesson 3: Amdahl's Law is Not Just Theory

Before this assignment, Amdahl's Law seemed like an abstract theoretical concept. Seeing it manifest in real performance data — where measured speedup (2.35x) closely matches the theoretical prediction (2.43x) — made the concept concrete. The sequential fraction (42%) directly determines the maximum achievable speedup, and no amount of optimization can overcome this fundamental limit without changing the algorithm itself.

Lesson 4: Verification is Critical

Parallel programs are notoriously difficult to debug because race conditions and synchronization errors can produce subtly incorrect results. Implementing multiple verification methods caught several bugs early: incorrect vertex range calculation causing some vertices to be skipped, visited array not being properly synchronized across processes, and distance updates using stale data from previous iterations.

5.3 What I Would Do Differently

Improvement 1: Implement Delta-Stepping Algorithm

The Delta-Stepping algorithm is a parallel variant that relaxes the strict sequential ordering requirement by processing vertices in “buckets” based on their distance ranges. Multiple vertices within the same bucket can be processed simultaneously, reducing synchronization frequency. This could potentially achieve 3–4x speedup with 8 processes instead of the current plateau at 2–3x.

Improvement 2: Use Asynchronous Communication

The current implementation uses blocking `MPI_Allreduce`. Using `MPI_Iallreduce` (non-blocking) would allow overlapping communication with computation, allowing processes to continue updating local distances while the reduction is in progress. This could reduce idle time and improve efficiency by 10–20%.

Improvement 3: Implement Priority Queue for Better Sequential Baseline

The current sequential implementation uses a linear search to find the minimum distance vertex, giving $O(V^2)$ complexity. Using a binary heap or Fibonacci heap would reduce this to $O((V+E) \log V)$, making the sequential version faster and providing a more realistic baseline for comparison.

Improvement 4: Test on True Distributed Hardware

All experiments were run on a single shared machine, meaning processes competed for the same CPU cores and memory bandwidth. Testing on a true cluster with dedicated cores per process, fast interconnect, and distributed memory would likely show better speedup. Expected speedup with 4 processes on proper hardware: 3.0–3.5x.

Improvement 5: Implement Hybrid MPI+OpenMP

A hybrid approach could combine MPI for inter-node communication (coarse-grained parallelism) with OpenMP within each node for shared-memory parallelism (fine-grained). Each MPI process would spawn multiple OpenMP threads to search for local minima, reducing the number of MPI processes and communication overhead while still utilizing all available cores.

5.4 Surprises

Surprise 1: Efficiency Varies Significantly with Problem Size

Efficiency was expected to be relatively consistent across problem sizes, but the data shows significant variation: $N=2,000$ at $P=4$ gives 58.8% efficiency, while $N=3,000$ at $P=4$ gives only 23.5%. The $N=3,000$ case is particularly surprising — it performs worse than both smaller and larger problem sizes, likely due

to cache effects (the 36MB adjacency matrix exceeding L3 cache), memory bandwidth saturation, and NUMA effects.

Surprise 2: Single Process MPI is Sometimes Slower Than Sequential

For several test cases, running the parallel version with $P=1$ is actually slower than the sequential version (speedup < 1.0). This is caused by MPI initialization overhead, MPI_Allreduce function call overhead even with one process, additional memory allocations for MPI buffers, and less effective compiler optimizations with MPI code.

Surprise 3: Communication Overhead Grows Linearly with Problem Size

Communication overhead was initially expected to be relatively constant, but the data shows it grows linearly with V because there are V iterations, each requiring one MPI_Allreduce. For $V=4,000$, there are 4,000 synchronization points, each taking ~ 0.001 s, totaling ~ 4 seconds of communication overhead.

Surprise 4: Graph Density Has Minimal Impact on Parallel Performance

Different edge densities (0.2, 0.3, 0.5) were tested, expecting denser graphs to show better speedup due to more computation per iteration, but the results were nearly identical. Since the bottleneck is MPI_Allreduce (not the distance update computation), parallelization benefits are limited regardless of graph structure.

6. Conclusion

This assignment successfully implemented a parallel version of Dijkstra's shortest path algorithm using MPI, achieving a maximum speedup of 2.35x with 4 processes on a graph with 2,000 vertices. The results clearly demonstrate the challenges of parallelizing iterative algorithms with global synchronization requirements.

Key Findings:

- Speedup is limited by Amdahl's Law due to 40–50% sequential fraction (communication)
- Efficiency decreases as process count increases (58.8% at $P=4$)
- Communication overhead grows linearly with problem size (V iterations $\times$ cost per sync)
- Measured performance closely matches theoretical predictions (2.35x vs. predicted 2.43x)

Practical Implications:

- Parallel Dijkstra is beneficial for medium-sized graphs (1,000–5,000 vertices) with 2–4 processes
- Beyond 4 processes, communication overhead outweighs computational benefits
- For very large graphs, alternative algorithms (Delta-Stepping, A^*) may be more suitable
- Hardware characteristics (cache size, memory bandwidth, network speed) significantly impact performance

Educational Value:

This assignment provided hands-on experience with MPI collective operations (Bcast, Allreduce, Gatherv), understanding Amdahl's Law through real performance data, recognizing when algorithms are inherently difficult to parallelize, trade-offs between memory usage and communication overhead, and the importance of verification and correctness testing in parallel programs.

7. References

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Appendix A: Compilation and Execution Commands

Sequential Version

```
cd sequential
gcc -o dijkstra sequential.c -lm -O2
./dijkstra -n 2000 -d 0.3 -s 0 -v
```

Parallel Version

```
cd parallel
mpicc -o dijkstra_parallel parallel.c -lm -O2
mpirun -np 4 ./dijkstra_parallel -n 2000 -d 0.3 -s 0 -v
```

Running Experiments

```
chmod +x run_experiments.sh
./run_experiments.sh
```

Generating Plots

```
cd results
python3 generate_plots.py
```

Verifying Correctness

```
chmod +x verify_correctness.sh
./verify_correctness.sh
```

Appendix B: Command Line Arguments

Argument	Description	Default	Example
-n <size>	Number of vertices in graph	100	-n 2000
-d <density>	Edge density (0.0 to 1.0)	0.3	-d 0.5
-s <source>	Source vertex index	0	-s 10
-v	Verbose output (print all distances)	Off	-v

Appendix C: Performance Summary Table

Metric	Value	Notes
Best Speedup	2.35x	N=2,000, P=4

Best Efficiency	97.0%	N=2,000, P=1
Worst Efficiency	23.5%	N=3,000, P=4
Communication Overhead	40–50%	Estimated from timing data
Optimal Process Count	2–4	Beyond this, overhead dominates
Largest Graph Tested	4,000 vertices	Limited by memory and time
Amdahl's Law Prediction	2.43x	Close match to measured 2.35x

End of Report